Exercises 3.9 3, 5, 17, 22 (1.01 m/s), 25, 32 (−0.25 rad/s), 45.

3. let x be the length of each side

A be the area of the square

$$A = x^2$$

$$\frac{dA}{dt} = 2x \cdot \frac{dx}{dt}$$

$$= 2\sqrt{16} \cdot 6$$

$$= 48$$

5. The radius of a spherical ball is increasing at a rate of 2 cm/min. At what rate is the surface area of the ball increasing when the radius is 8 cm?

$\frac{dr}{dt}$ $\frac{dA}{dt}$

$\frac{dr}{dt} = 2$ $\qquad \frac{dS}{dt}\Big|_{r=8}$

$S = 4\pi r^2$

$\frac{dS}{dt} = 4\pi \cdot 2r \cdot \frac{dr}{dt}$

$\frac{dS}{dt} = 8\pi \times 8 \times 2$

$= 402.12$

11. The weight w of an astronaut (in pounds) is related to her height h above the surface of the earth (in miles) by

$$w = w_0\left(\frac{3960}{3960 + h}\right)^2$$

where w_0 is the weight of the astronaut on the surface of the earth. If the astronaut weighs 130 pounds on earth and is in a rocket, being propelled upward at a speed of 12 mi/s, find the rate at which her weight is changing (in lb/s) when she is 40 miles above the earth's surface.

$\frac{dh}{dt} = 12$

$h = 40$

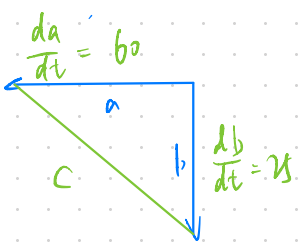
$$\frac{dw}{dt} = w_0 \cdot 2\left(\frac{3960}{3960+h}\right) \cdot \frac{-3960 \cdot \frac{dh}{dt}}{(3960+h)^2}$$

$$= 130 \times 2 \cdot \frac{3960}{3960+40} \cdot \frac{-3960 \times 12}{(3960+40)^2}$$

$$= -0.7645$$

Exercises 3.9 3, 5, 17, 22 (1.01 m/s), 25, 32 (−0.25 rad/s), 45.

17. Two cars start moving from the same point. One travels south at 60 mi/h and the other travels west at 25 mi/h. At what rate is the distance between the cars increasing two hours later?



let a be the distance traveling west from the starting point

b ... ... south ...

c ... ... between 2 cars

$$c^2 = a^2 + b^2$$

$$2c \cdot \frac{dc}{dt} = 2a \cdot \frac{da}{dt} + 2b \cdot \frac{db}{dt}$$

after 2 h:

$a = 60 \times 2 = 120$

$b = 25 \times 2 = 50$

$c = \sqrt{a^2 + b^2} = \sqrt{120^2 + 50^2} = 130$

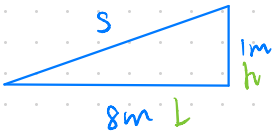
$$2 \times 130 \frac{dc}{dt} = 2 \times 120 \times 60 + 2 \times 50 \times 25$$

$$\frac{dc}{dt} = 65$$

Exercises 3.9 3, 5, 17, 22 (1.01 m/s), 25, 32 (−0.25 rad/s), 45.

22. A boat is pulled into a dock by a rope attached to the bow of the boat and passing through a pulley on the dock that is 1 m higher than the bow of the boat. If the rope is pulled in at a rate of 1 m/s, how fast is the boat approaching the dock when it is 8 m from the dock?





$s^2 = L^2 + h^2$ (1)

$2s\frac{ds}{dt} = 2L\frac{dL}{dt} + 2h\frac{dh}{dt}$

$2\sqrt{8^2+1}(-1) = 2\times 8\cdot\frac{dL}{dt} + 2\times 1\times 0$

$-16.12 = 16\frac{dL}{dt} + 2$

$\frac{dL}{dt} = -0.88$

$s^2 = L^2 + 1$

$2s\frac{ds}{dt} = 2L\frac{dL}{dt}$

$\frac{dL}{dt} = \frac{s}{L}\cdot\frac{ds}{dt}$

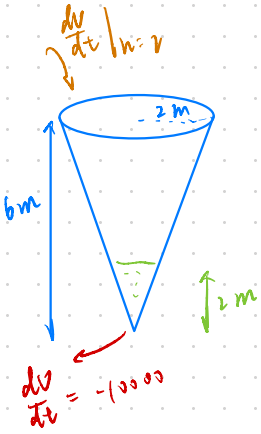
$= \frac{\sqrt{8^2+1}}{8}\times(-1) = -1.01$ m/s

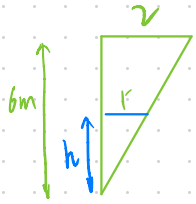
25. Water is leaking out of an inverted conical tank at a rate of 10,000 cm^3/min at the same time that water is being pumped into the tank at a constant rate. The tank has height 6 m and the diameter at the top is 4 m. If the water level is rising at a rate of 20 cm/min when the height of the water is 2 m, find the rate at which water is being pumped into the tank.

$\frac{dV}{dt} = -10000$

$\frac{dh}{dt} = 20$

200 cm





$\frac{r}{h} = \frac{2}{6}$

$r = \frac{h}{3}$

let V be the volumn of the water

r ... reading

C be the rate at which water is pumped in

$\frac{dV}{dt} = C - 10000$

$V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi \left(\frac{h}{3}\right)^2 h$

$= \frac{\pi}{27} h^3$

$\frac{dV}{dt} = \frac{\pi}{9} h^2 \cdot \frac{dh}{dt}$

$= \frac{\pi}{9} \times (200)^2 \times 20$

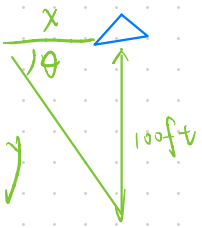
$= 27925.68$

$C = \frac{dV}{dt} + 10000$

$= 37925.68$

32. A kite 100 ft above the ground moves horizontally at a speed of 8 ft/s. At what rate is the angle between the string and the horizontal decreasing when 200 ft of string has been let out?

$\frac{d\theta}{dt}\Big|_{y=200}$



$\cos\theta = \frac{x}{y}$

$\theta = \cos^{-1}\left(\frac{x}{y}\right)$

$\theta = \cos^{-1}\left(\frac{x}{y}\right)$

$x = \sqrt{y^2 - 100^2}$

$= \sqrt{200^2 - 100^2}$

$= 173.21$

$\tan\theta = \frac{100}{x}$

$x = \frac{100}{\tan\theta}$

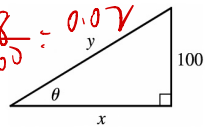
$\frac{dx}{dt} = \frac{-100\sec^2\theta \cdot \frac{d\theta}{dt}}{\tan^2\theta}$

$= -100 \cdot \frac{\cos^2\theta}{\sin^2\theta} \cdot \frac{d\theta}{dt}$

$\frac{d\theta}{dt} = \cos$ 用含θ的表达式把它得到 $= -\frac{100}{\sin^2\theta}\frac{d\theta}{dt}$

$\frac{d\theta}{dt} = -\frac{\sin^2\theta}{100} \cdot \frac{dx}{dt}$

$= -\frac{8}{400} = 0.02$



如用cos我有sin会引入变量y且$\frac{dy}{dt}$未知

32. We are given $dx/dt = 8$ ft/s. $\cot\theta = \dfrac{x}{100} \Rightarrow x = 100\cot\theta \Rightarrow$

$\dfrac{dx}{dt} = -100\csc^2\theta\,\dfrac{d\theta}{dt} \Rightarrow \dfrac{d\theta}{dt} = -\dfrac{\sin^2\theta}{100}\cdot 8$. When $y = 200$, $\sin\theta = \dfrac{100}{200} = \dfrac{1}{2} \Rightarrow$

$\dfrac{d\theta}{dt} = -\dfrac{(1/2)^2}{100}\cdot 8 = -\dfrac{1}{50}$ rad/s. The angle is decreasing at a rate of $\frac{1}{50}$ rad/s.

$\csc\theta = \frac{1}{\sin\theta}$

$\sin\theta = \frac{100}{200} = \frac{1}{2}$

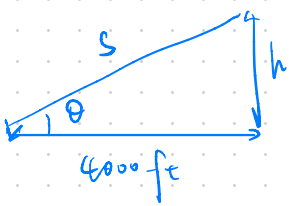
$-\frac{1}{4} \times \frac{1}{100} \times 8$

45. A television camera is positioned 4000 ft from the base of a rocket launching pad. The angle of elevation of the camera has to change at the correct rate in order to keep the rocket in sight. Also, the mechanism for focusing the camera has to take into account the increasing distance from the camera to the rising rocket. Let's assume the rocket rises vertically and its speed is 600 ft/s when it has risen 3000 ft.

$\frac{dh}{dt}\Big|_{h=3000} = 600$

(a) How fast is the distance from the television camera to the rocket changing at that moment? $\frac{ds}{dt}\Big|_{h=3000}$

(b) If the television camera is always kept aimed at the rocket, how fast is the camera's angle of elevation changing at that same moment?



a)

$$s^2 = 4000^2 + h^2$$

$$2s\frac{ds}{dt} = 2h\cdot\frac{dh}{dt}$$

$$s = \sqrt{4000^2 + 3000^2} = 5000$$

$$2\times 5000\times\frac{ds}{dt} = 2\times 3000\times 600$$

$$\frac{ds}{dt} = 360$$

b)

$$\tan\theta = \frac{h}{4000}$$

$$\sec^2\theta\cdot\frac{d\theta}{dt} = \frac{4000\cdot\frac{dh}{dt}}{4000^2}\qquad \frac{1}{4000}\cdot\frac{dh}{dt}$$

$$\left(\frac{5}{4}\right)^2\frac{d\theta}{dt} = \frac{4000\times 600}{4000^2}$$

$$\frac{d\theta}{dt} = 0.12\times 0.096$$